

Ex. 3: 5G NR Throughput for QPSK–256QAM

Code:

```
clc;
clear;
close all;
BW = 100e6;
subcarrierSpacing = 30e3;
Nsub = floor(BW / subcarrierSpacing);
codeRate = 0.85;
MIMO_layers = 4;
mod_scheme = {'QPSK','16QAM','64QAM','256QAM'};
bits_per_symbol = [2 4 6 8];
throughput = zeros(1,length(bits_per_symbol));
for i = 1:length(bits_per_symbol)
    bps = bits_per_symbol(i);
    T = bps * Nsub * codeRate * MIMO_layers;
    throughput(i) = T / 1e6;
end
disp('5G NR Throughput for Different Modulation Schemes');
disp('-----');
for i = 1:length(mod_scheme)
    fprintf('%-7s : %.4f Mbps\n', mod_scheme{i}, throughput(i));
end
figure;
bar(throughput);
set(gca,...
'XTick',1:length(mod_scheme),...
'XTickLabel',mod_scheme,...
'FontSize',12,...
'FontWeight','bold');
xlabel('Modulation Scheme');
ylabel('Throughput (Mbps)');
title('5G NR Throughput for Different Modulation Schemes');
grid on;
```

